RAG Evaluation

Interviewer:

I have seen your RAG chatbot. It gives impressive answers in your demo. But how do you prove it's production-ready and not just demo-ready?

Candidate:

A good-looking demo doesn't necessarily mean the RAG system is production-ready. To prove that it's reliable, we perform **RAG Evaluations (RAG Evals)**.

A RAG system has **two main responsibilities**:

1. It should **retrieve the correct information** from the knowledge base.
2. It should **generate the correct answer** using that retrieved information.

If either of these fails, the user can get an incorrect answer. That's why we evaluate both retrieval quality and generation quality separately.

Interviewer:

Can you explain that with a real example?

Candidate:

Sure.

Let's take an airline chatbot.

A passenger asks:

"My flight is delayed by 6 hours. Am I eligible for free hotel accommodation?"

The airline policy says:

"Hotel accommodation is provided only if the delay requires an overnight stay and the delay is caused by the airline."

Now my RAG system has to answer two questions.

First,

Did it retrieve the correct airline policy?

Second,

Did the LLM correctly understand that policy and generate the right answer?

Both retrieval and generation have to work correctly.

Interviewer:

How do you evaluate retrieval?

Candidate:

Suppose my retriever returns the **Top 5 documents**.

I mainly use two metrics.

1. Recall@K

Recall@K checks whether the correct document is present in the Top-K retrieved documents.

For example, if the correct airline policy is present in the Top 5 results, then Recall@5 is successful.

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2. Precision@K

Precision@K measures how many of the retrieved documents are actually relevant.

For example, if only 3 out of the 5 retrieved documents are relevant, then Precision@5 is 60%.

A retriever should not return many irrelevant documents just to include one correct document.

So both **high Recall** and **high Precision** indicate good retrieval quality.

Interviewer:

Suppose the correct document was retrieved. Can the answer still be wrong?

Candidate:

Yes, absolutely.

For example, the LLM might answer:

"Yes, you are eligible because your flight is delayed by 6 hours."

But that's actually incorrect.

The airline policy clearly says the hotel is provided **only if the delay requires an overnight stay and the delay is caused by the airline.**

So in this case:

- Retrieval was correct.
- Generation failed.

That's why evaluating retrieval alone is not enough. We also need to evaluate the generated answer.

Interviewer:

How do you evaluate the generated answer?

Candidate:

I mainly use three important metrics.

1. Faithfulness

Faithfulness checks whether every statement generated by the LLM is supported by the retrieved documents.

The model should not invent any information.

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2. Answer Relevance

This checks whether the generated answer actually answers the user's question.

Sometimes the answer is correct but doesn't directly answer what the user asked.

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3. Correctness

Correctness checks whether the final answer is factually correct when compared with the expected answer.

So even if the answer sounds fluent, it should still be factually accurate.

Interviewer:

Where do RAG Evals help in real engineering?

Candidate:

They help us compare different versions of the RAG pipeline using objective metrics.

For example,

Version 1

- Chunk Size = 500
- Top-K = 5
- Embedding Model = Model A

Later, I make some improvements.

Version 2

- Change the chunking strategy
- Add a Cohere Reranker
- Switch to a better embedding model

Without RAG Evals, I can only say,

"Version 2 feels better."

But that's just my opinion.

With RAG Evals, I can actually measure the improvement.

For example,

- Recall@5 improves from **70% to 90%**
- Precision improves
- Faithfulness improves
- Correctness improves
- Hallucinations reduce

Now my engineering decisions are based on measurable data instead of assumptions.

Interviewer:

Can you explain RAG Evals in one sentence?

Candidate:

Sure.

RAG Evals are a set of evaluation metrics that measure how well a RAG system retrieves the correct information and generates accurate, faithful, and relevant answers. They help us improve the system using data-driven metrics instead of guesswork.